



Introduction to PSOC™ Edge E84

Getting Started with the Development Ecosystem (E2)



Agenda

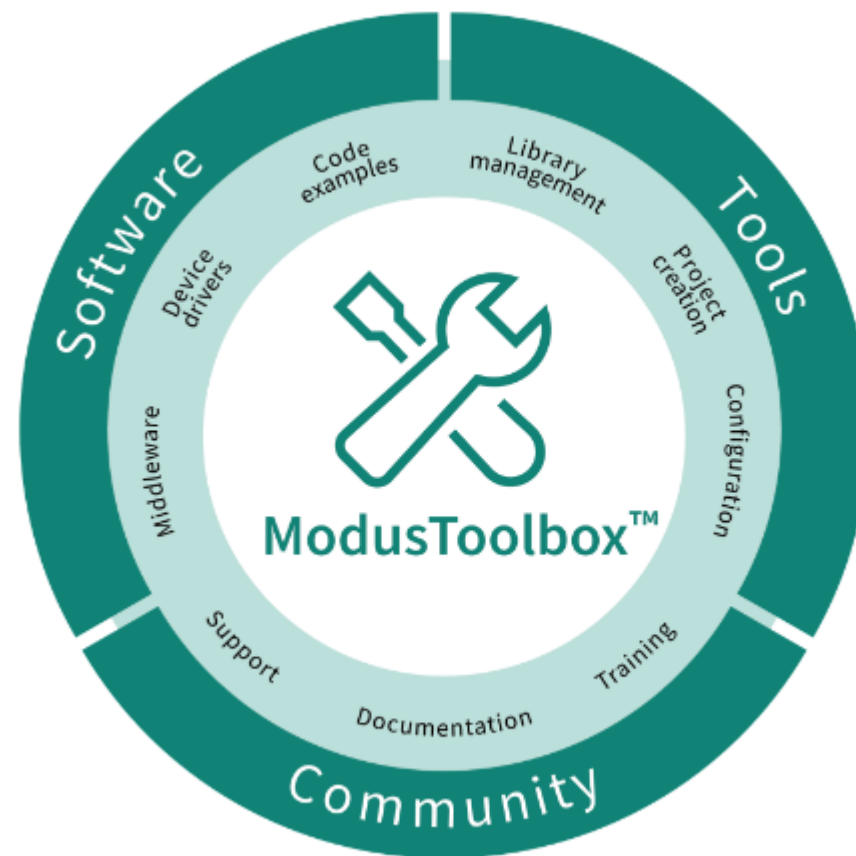
- ❑ Development ecosystem for PSOC™ Edge E84
 - ❑ Resources (Documentation / Application Notes)
 - ❑ Firmware / Application Development (ModusToolbox™, DEEPCRAFT™)
- ❑ Code Example #1: Hello World + Blinky
- ❑ Code Example #2: IPC Semaphores using PDL API

PSOC™ Edge E84 Resources

- ☐ PSOC™ Edge E84 MCU
- ☐ Datasheets
- ☐ Reference manuals (RM): architecture and registers
- ☐ Application notes
- ☐ Code examples
- ☐ Development kits
- ☐ KBAs and community forum

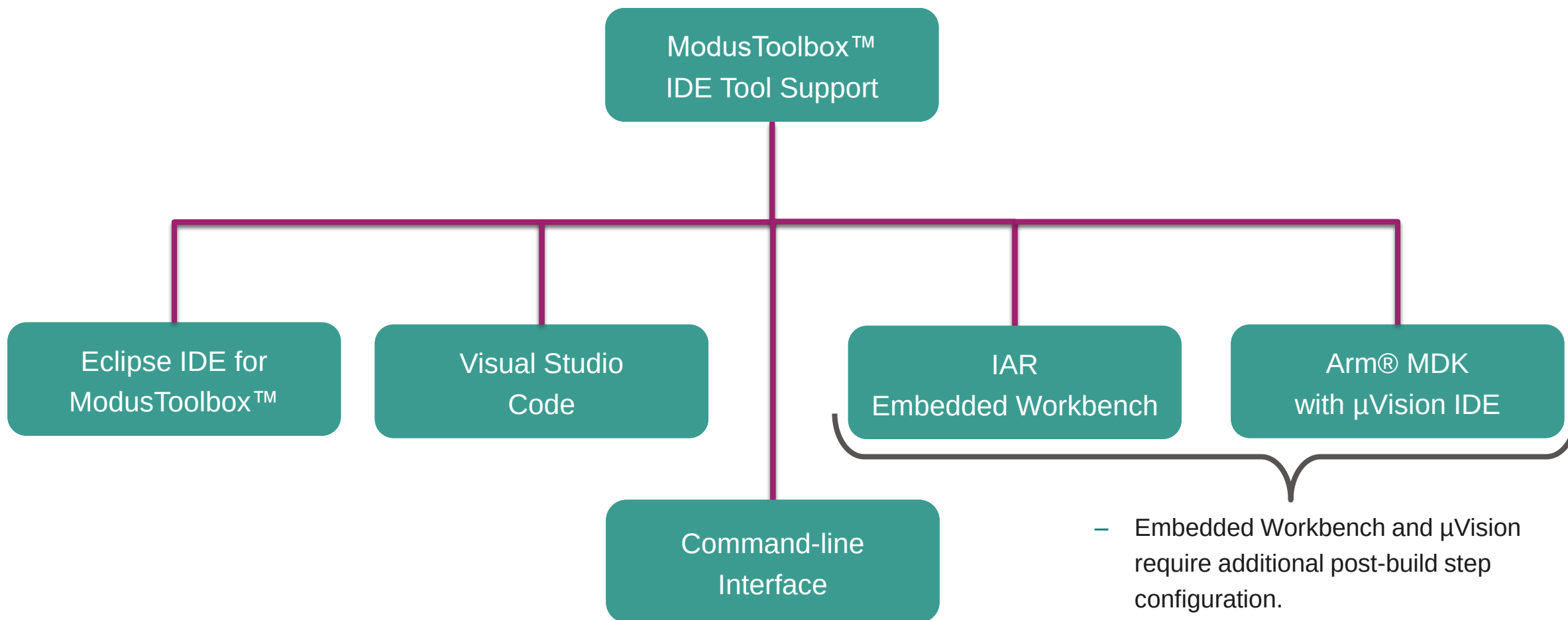
Firmware / Application Development

- ❑ **ModusToolbox™ provides**
 - ❑ **Development environment for Infineon MCUs**
 - ❑ **Flexible set of tools for project and system level configuration**
 - ❑ **Low-level drivers**
 - ❑ **Libraries**
 - ❑ **Operating system support (Linux, macOS, Windows)**



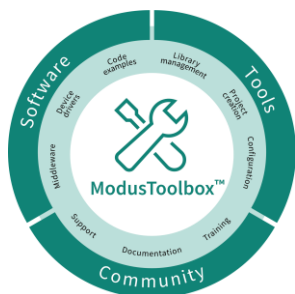
Reference link : <https://www.infineon.com/ModusToolboxInstallguide>

Support for IDEs



- Embedded Workbench and μVision require additional post-build step configuration.
- Details documented within IDE respective user manuals

Understanding ModusToolbox™ and Zephyr



- Provides a **targeted ecosystem** focused on development of **Infineon devices**.

- IDE support, including **Eclipse IDE**, **VS Code**, and **3rd party IDEs**.
- Support for GCC, IAR, and Arm compilers using ‘make’ based **ModusToolbox™ build system**.
- Graphical **UI configurations** tool
- Leverages **Infineon developed** low-level drivers, middleware abstraction, middleware, and code examples.
- Enables key partner middleware via **ModusToolbox™ & Friends** program

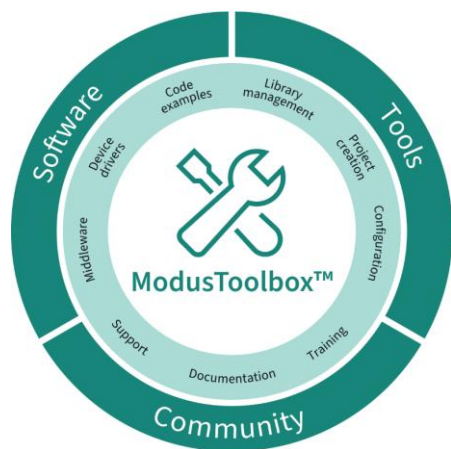


Zephyr®

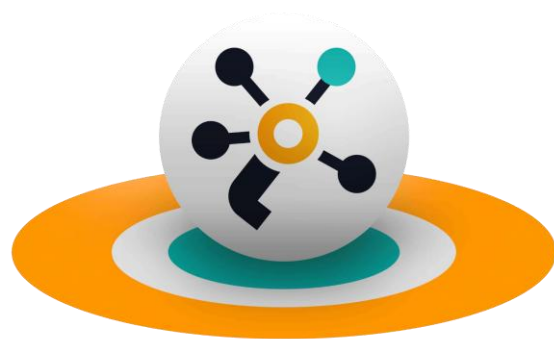
- Provides a **community driven ecosystem** with support for a targeted set of MCUs from a **wide range of silicon providers**.

- Targeted at **command-line users** with some community support for **VS Code**.
- Default support for GCC compiler using ‘west’ based **Zephyr build environment**. Additional toolchains are now being supported.
- Text-based configuration via **devicetree and Kconfig**
- Leverages vendor provided device support; Infineon support based on **low-level drivers**.
- **Community contributed** middleware, with support for Zephyr defined driver API, stacks, and RTOS.

Overview of DEEPCRAFT™ Studio and ModusToolbox™ workflow



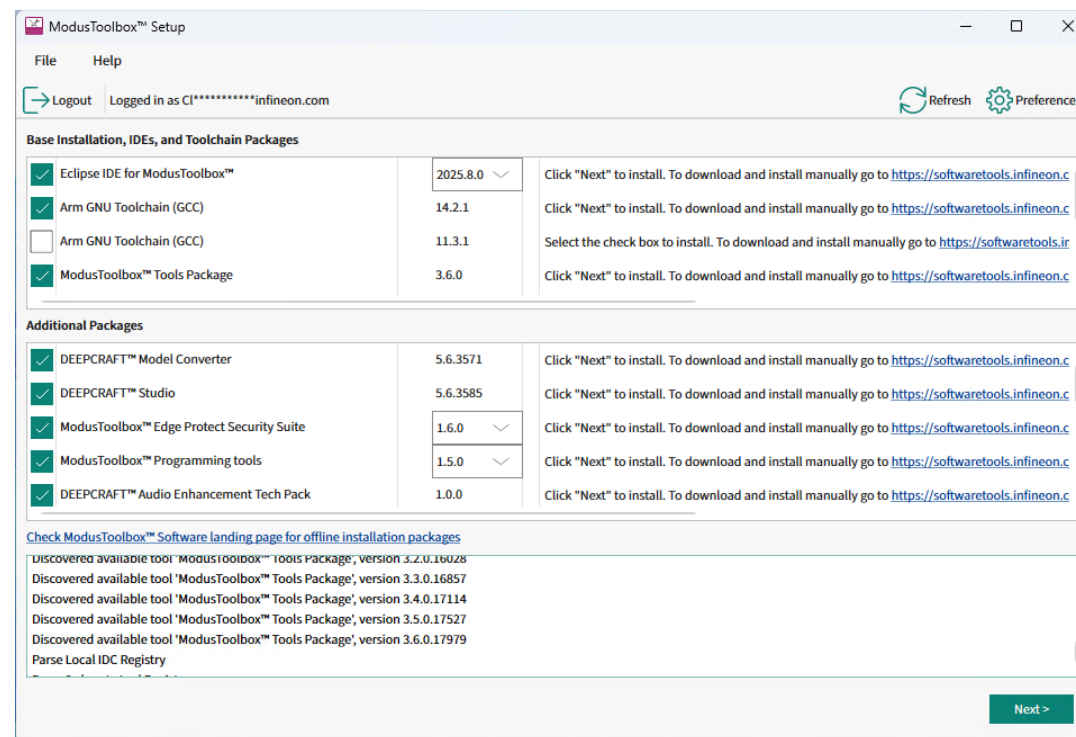
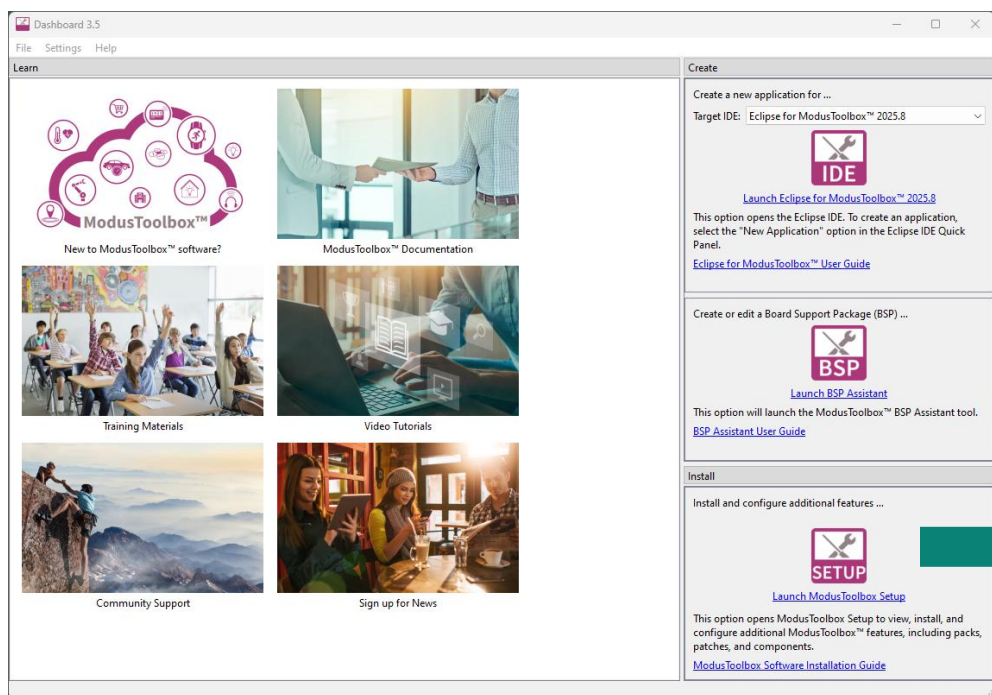
- Installation of ModusToolbox™ and DEEPCRAFT™ supported within the [ModusToolbox™ Setup](#) program
- Code Examples aligned with DEEPCRAFT™ Studio [Accelerator projects](#) for IMU, Vibration, and Microphone
- Code Examples aligned with DEEPCRAFT™ Studio [Data Collection with Graph UX](#)
- Code Examples aligned with DEEPCRAFT™ Model Converter [Profiling](#)
- [Project connector](#) files in ModusToolbox™ applications with Quick Panel shortcuts for linking to DEEPCRAFT™ Studio Projects



- Integration of ModusToolbox™ Project Creator, supporting coordinated DEEPCRAFT™ Studio and ModusToolbox™ project creation
- Integration ModusToolbox™ Machine Learning Core Tools (ml-coretools), leveraged during Infineon device specific code generations and optimization
- ModusToolbox™ project detection and selection during model code generation

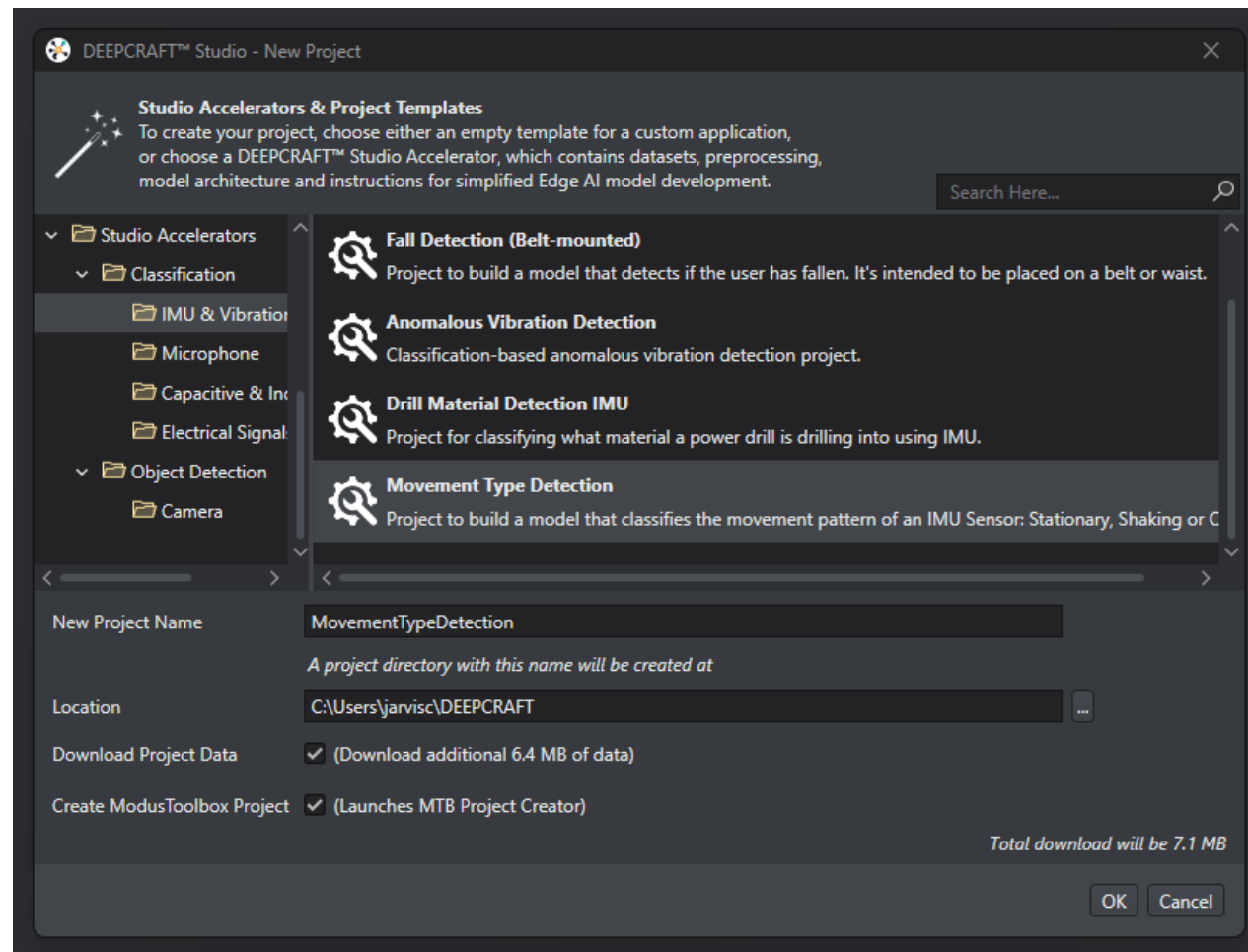
Installation support within ModusToolbox Setup

- ModusToolbox™ Setup program can be used to install and manage DEEPCRAFT™ applications, including Studio, Model Converter, and Audio Enhancement Tech Pack



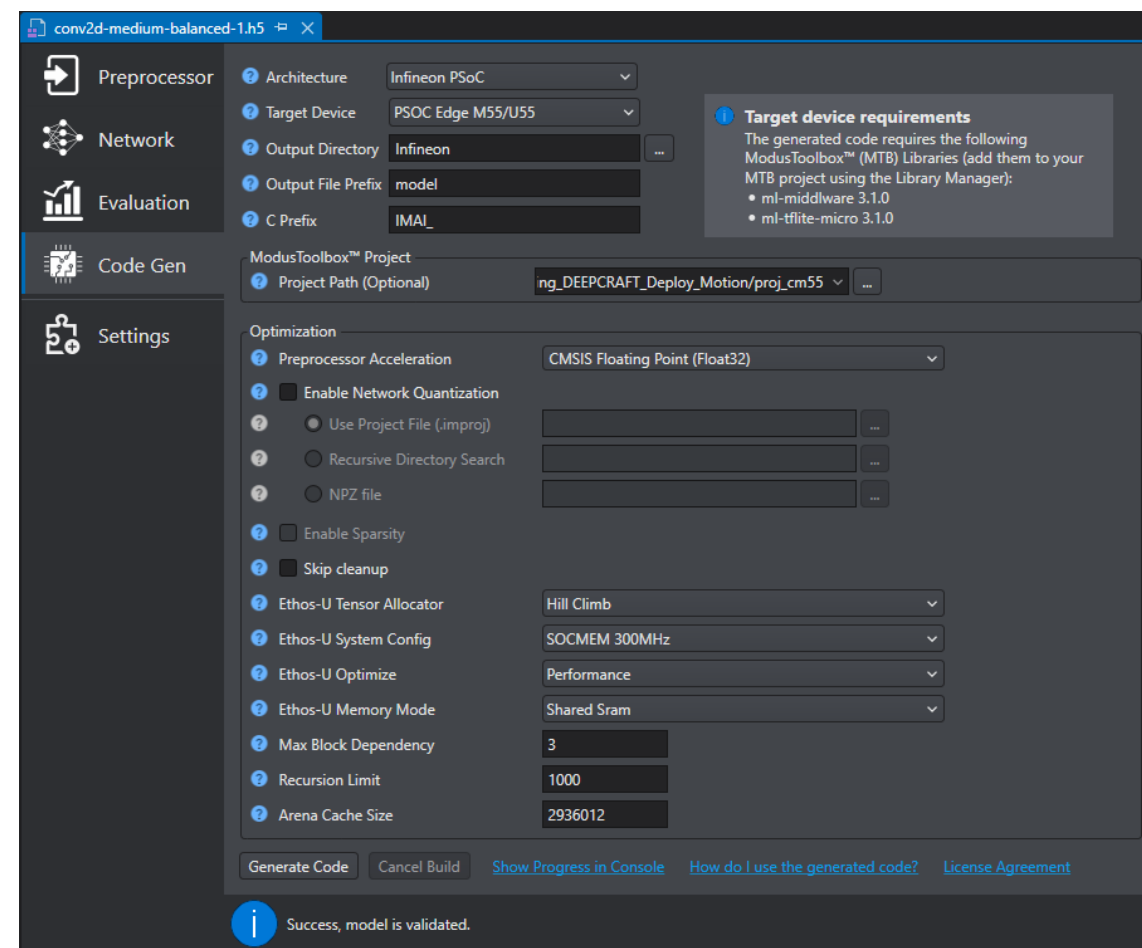
Code Examples with DEEPCRAFT™ and ModusToolbox™

- When creating a new DEEPCRAFT™ Studio project, a ModusToolbox™ project can be created at the same time.
- DEEPCRAFT™ Studio includes Studio Accelerator projects that align with ModusToolbox™ code examples, enabling models generated from Studio to be more easily leveraged within a ModusToolbox™ application.
- ModusToolbox™ code examples include DEEPCRAFT™ Deploy Audio and Deploy Motions applications.



DEEPCRAFT™ Studio Model Code Generation

- DEEPCRAFT™ Studio code generation dialog includes pre-defined architecture selections for Infineon devices, including PSOC™ Edge E84
- DEEPCRAFT™ Studio code generation integrates the ModusToolbox™ Machine Learning Core Tools (ml-coretool) with integrated Vela compiler supporting PSOC™ Edge E84.
- Selection of Infineon device enables project selection of the connected ModusToolbox™ project and core specific project path.
- Selected ModusToolbox™ project path will be used during the code generation step to enable iterative model development and updates.



Tools utilized in this presentation

- ModusToolbox™ v3.6
- Eclipse IDE for ModusToolbox™ v2025.4.0
- Edge Protect Security Suite v1.6
- ModusToolbox™ Programming Tools v1.5.0
- KIT_PSE84_EVAL_EPC2 BSP v1.0.0
- Terminal Emulator
- KIT_PSE84_EVAL revG

Key documentation leveraged in this video

- **AN235935** – Getting started with PSOC™ Edge E84 on ModusToolbox™ software
- ModusToolbox™ PSE8xxGP Device Support Library (DSL)

Application Booting sequence

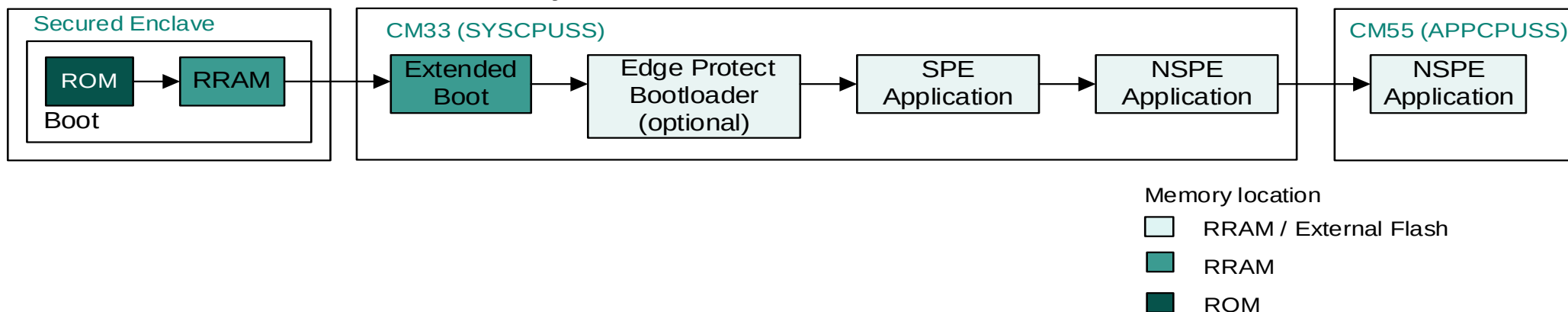
❑ Boot starts from Secured Enclave as the Root of Trust

❑ Off-the-shelf extended boot software

- Boot from RRAM or external flash
- Alternate serial interface for recovery boot
- SRAM loading for enhanced security

❑ Optional MCUboot based Edge Protect Bootloader

❑ One time revocable OEM RoT Key



Device lifecycle and provisioning

❑ Development

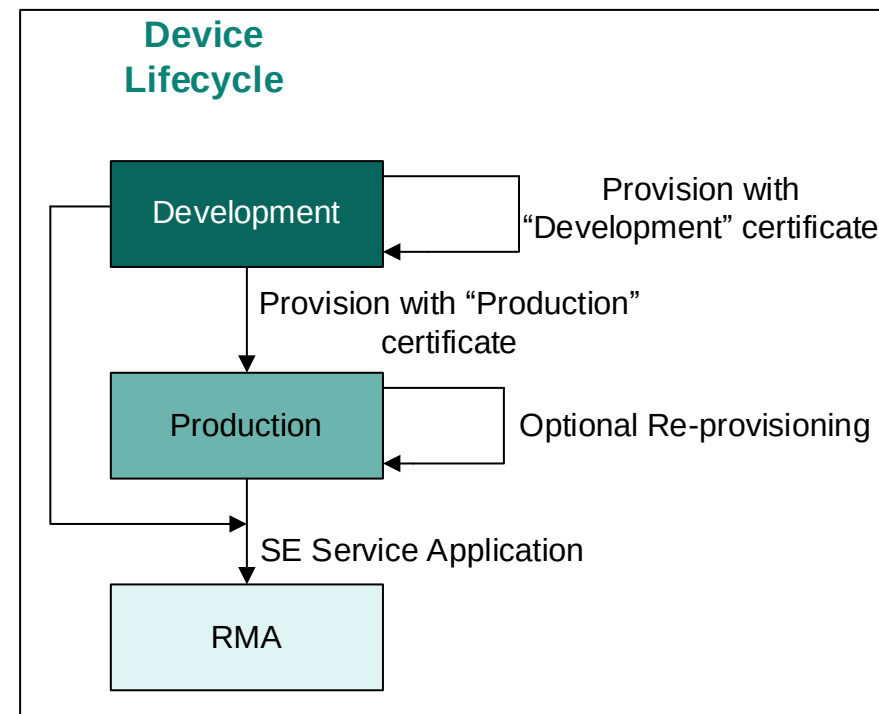
- Shipped from Infineon
- Provision with OEM certificate and policy
- For OEM development; until production

❑ Production

- One-time provisioning with “Production” certificate
- Re-provisioning if allowed by policy

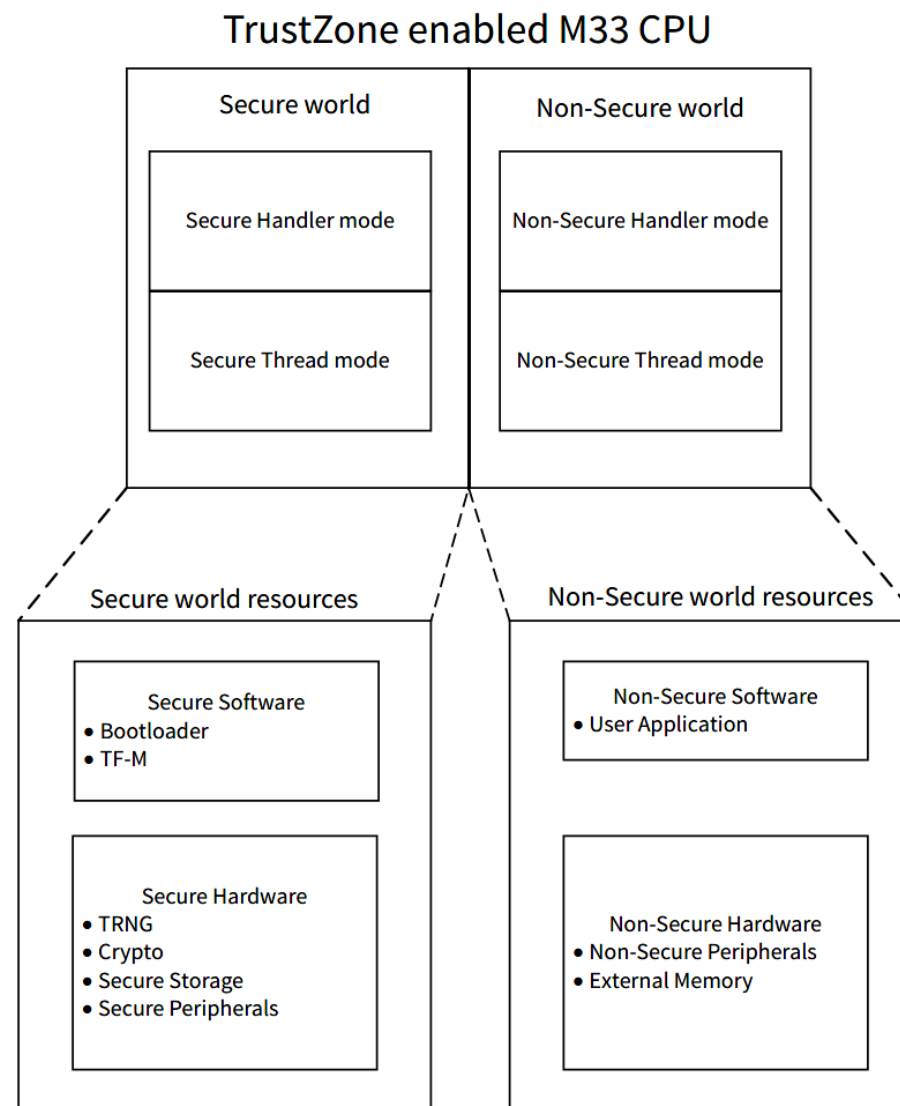
❑ RMA

- Transition from Development or Production
- Performed by Infineon signed SE Service Application
- Devices in RMA mode can be shipped to Infineon for field failure analysis



Secure and Non-Secure Operating Modes

- Arm® TrustZone- enabled Cortex® -M33 CPU
- Secure and non-secure CPU states
- Attribution units - Secure Attribution Unit (SAU) and Implementation Defined Attribution Unit (IDAU)
- Secure and non-secure aliasing of the memory map
- Secure and non-secure interrupts
- Banked resources between secure and non-secure states
- Secure and non-secure peripherals
- Secure and non-secure debug



Security Domain Assignment

C:/Users/jarvisc/mtw_e84/PSoc_Edge_Empty_Application/bsps/TARGET_APP_KIT_PSOCE84_EVK/config/design.modus - Device Configurator 4.10

File Edit Settings View Help

PSE846GPS2DBZC4 CYW55513IUBG

Peripherals Pins Analog-Routing System Peripheral-Groups Peripheral-Clocks DMA

Enter filter text...

Resource	Name(s)	Personality	Security Domain
✓ Low-Power Comparator 0	lpcomp_0_comp_0	Low Power Comparator	NSPE Domain
□ Low-Power Comparator 1	lpcomp_0_comp_1		NSPE Domain
Communication			
> Controller Area Network FD (CAN FD) 0			
> Quad Serial Memory Interface (QSPI) 0			
□ SD Host Controller (SDHC) 0	sdhc_0		NSPE Domain
□ SD Host Controller (SDHC) 1	sdhc_1		NSPE Domain
□ Serial Communication Block (SCB) 0	scb_0		
✓ Serial Communication Block (SCB) 1	scb_1	UART-3.0	NSPE Domain
□ Serial Communication Block (SCB) 2	scb_2		NSPE Domain
□ Serial Communication Block (SCB) 3	scb_3		SPE Domain
□ Serial Communication Block (SCB) 4	scb_4		NSPE Domain
□ Serial Communication Block (SCB) 5	scb_5		NSPE Domain
□ Serial Communication Block (SCB) 6	scb_6		SPE Domain
□ Serial Communication Block (SCB) 7	scb_7		NSPE Domain
□ Serial Communication Block (SCB) 8	scb_8		NSPE Domain
□ Serial Communication Block (SCB) 9	scb_9		NSPE Domain
□ Serial Communication Block (SCB) 10	scb_10		NSPE Domain
□ Serial Communication Block (SCB) 11	scb_11		NSPE Domain
□ TDM0	tdm_0_tdm_0		NSPE Domain
□ TDM1	tdm_0_tdm_1		NSPE Domain
□ Universal Serial Bus (USB) 0	usbhs_0		NSPE Domain
Digital			
System			

Serial Communication Block (SCB) 1 - Parameters

Enter filter text...

Name	Value
Overview	
? Configuration Help	Open UART (SCB) Documentation
General	
? Com Mode	Standard
? Baud Rate (bps)	115200
? Oversample	8
? Bit Order	LSB First
? Data Width	8 bits
? Parity	None
? Stop Bits	1 bit
? Enable Digital Filter	☐
Support RS-485	
? TX-Enable	☐
Flow Control	
? Enable Flow Control	☐
? CTS Polarity	Active Low
? RTS Polarity	Active Low
? RTS Activation Level	63
Connections	
? Clock	16 bit Divider 0 clk [USED]
? RX	P9[3] digital_inout [USED]
? TX	P9[2] digital_inout [USED]
? RX Trigger Output	<unassigne
? TX Trigger Output	<unassigne
Actual Baud Rate	
? Actual Baud Rate (bps)	115207
? Baud Rate Accuracy (%)	0.006
? Clock Frequency	121.650 kHz

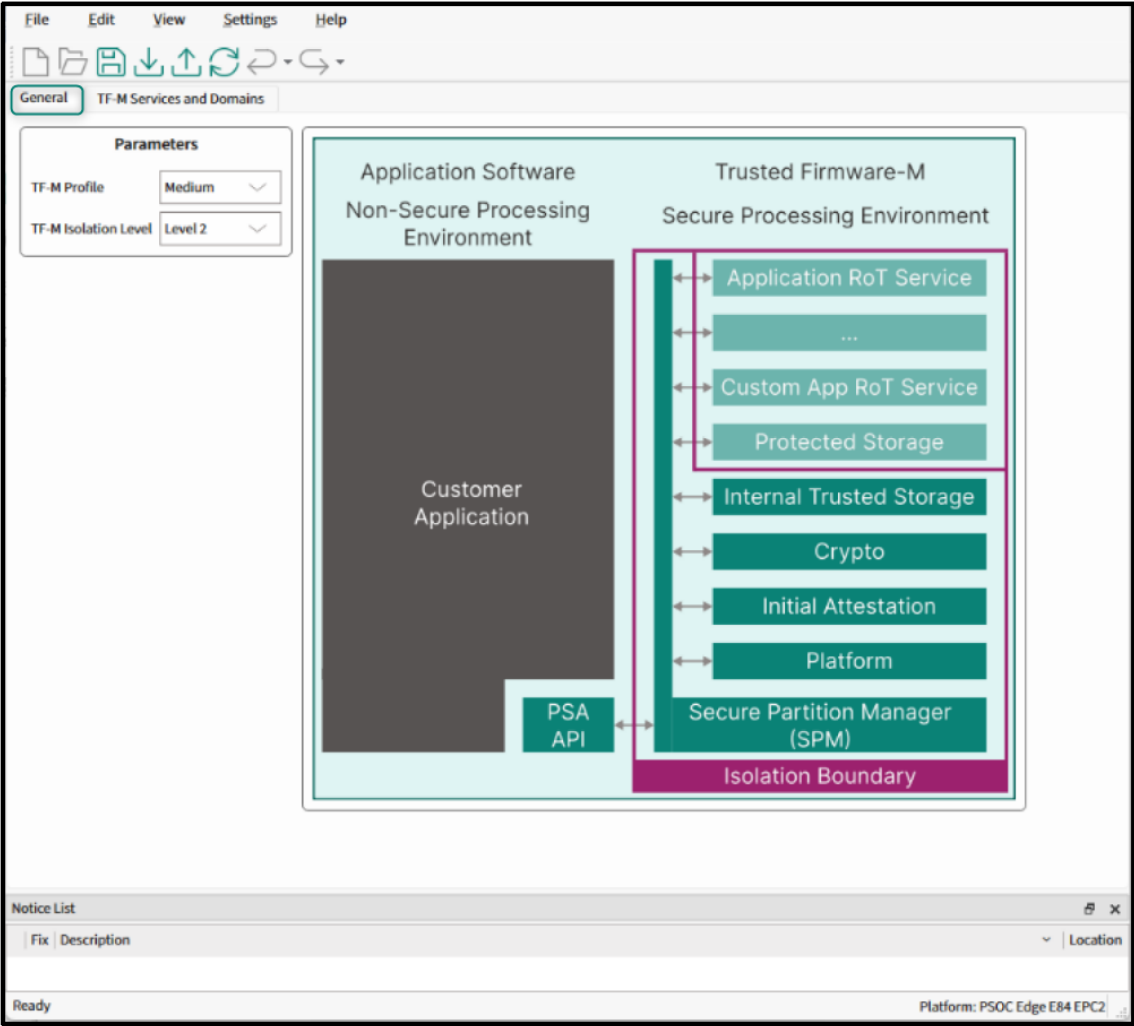
Serial Communication Block (SCB) 1 - Parameters Code Preview

Notice List

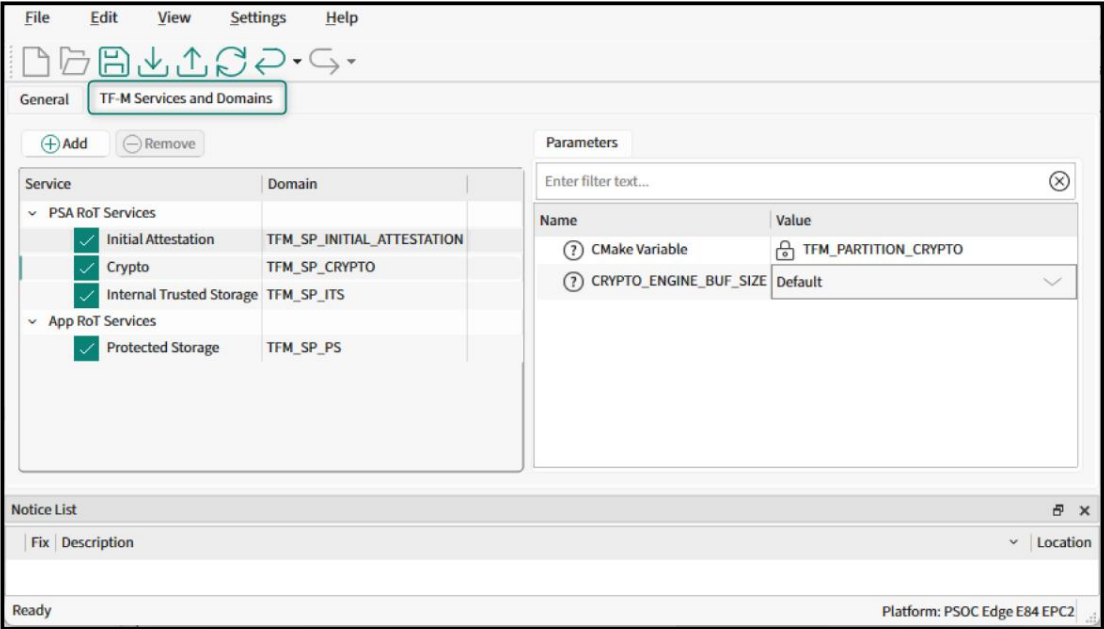
0 Errors 0 Warnings 0 Tasks 3 Infos

Fix	Description	Location
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11 Nov 2024 Ready



General tab

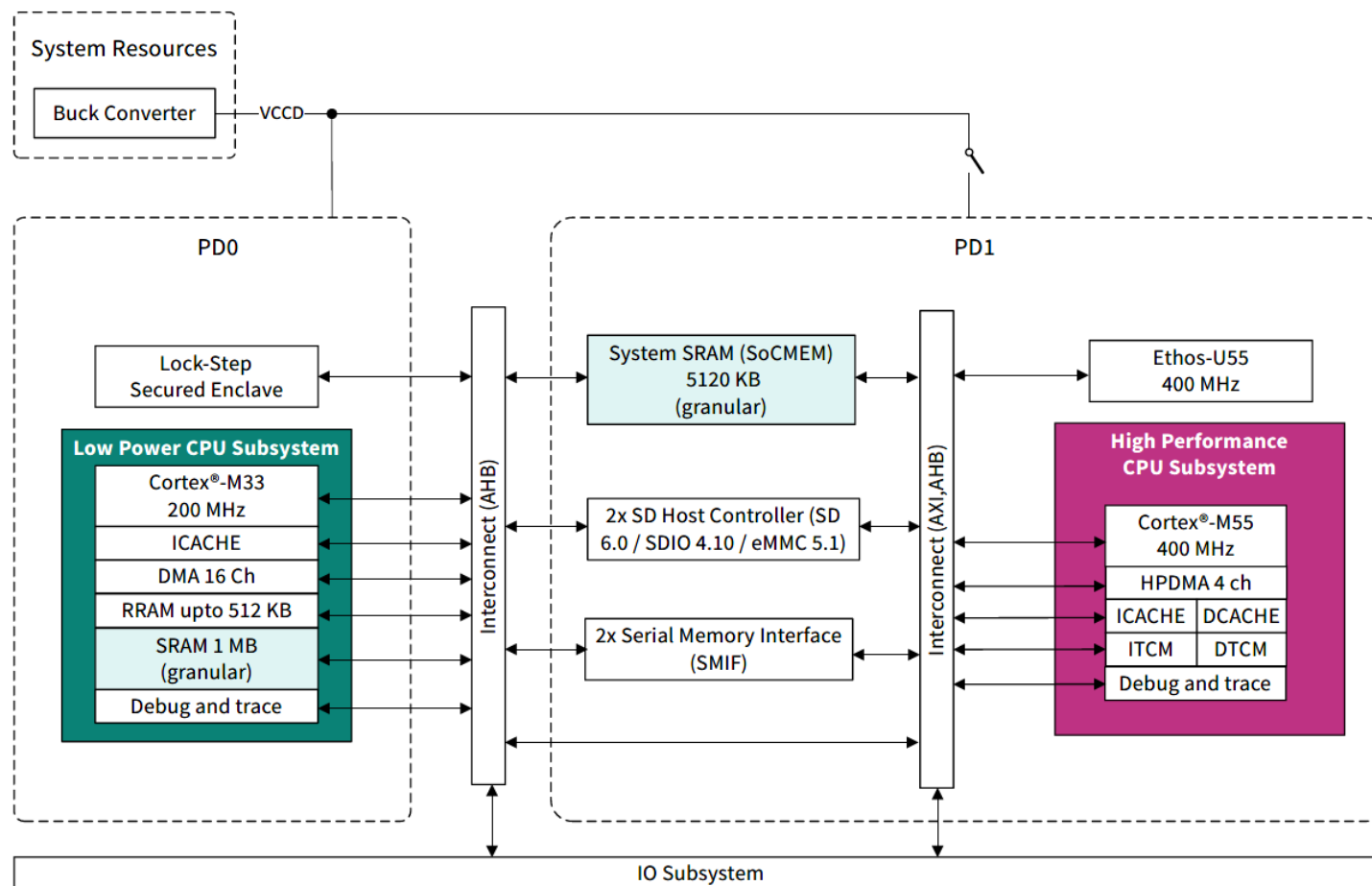


TF-M Services and Domains tab

Memory Overview

- 2 CPU Subsystem
- Each CPU equipped with cache
- 2 SRAM sections (1MB)
- System SRAM (5120KB)
- RRAM (512KB)*
- ITCM and DTCM (256KB each)
- SMIF0/1 (XIP capable) for external flash or RAM

* Some memory is reserved for extended boot and the secure application

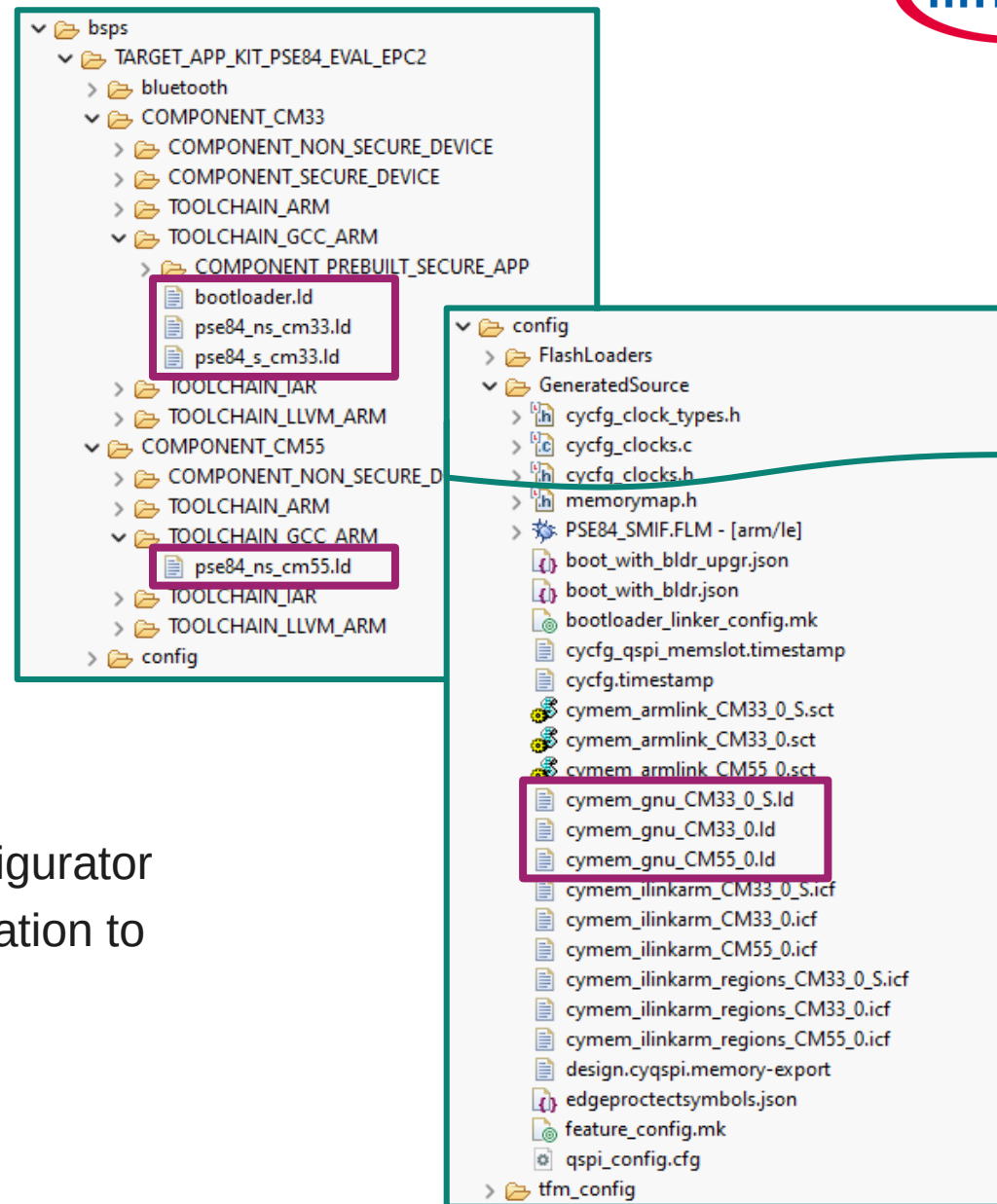


Linker Scripts

- Four top-level linker files
 - Bootloader
 - CM33 Secure
 - CM33 Nonsecure
 - CM55

- Three generated memory include files
 - CM33 Secure
 - CM33 Nonsecure
 - CM55

- Generated memory include files come from Device Configurator
- New user defined memory regions require direct modification to top-level linker file to leverage generated content

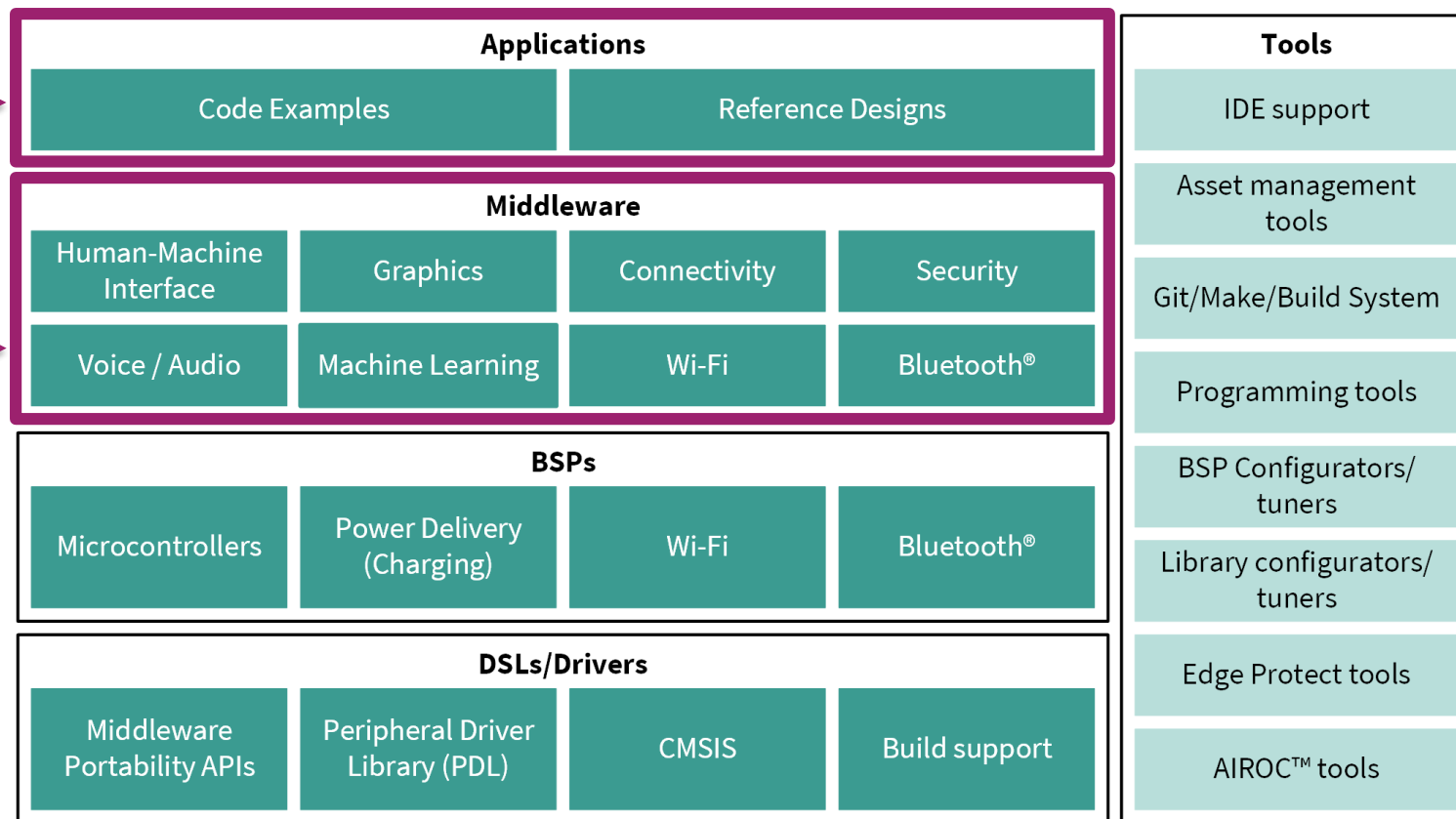


Firmware/application development :Software Resources

– Applications

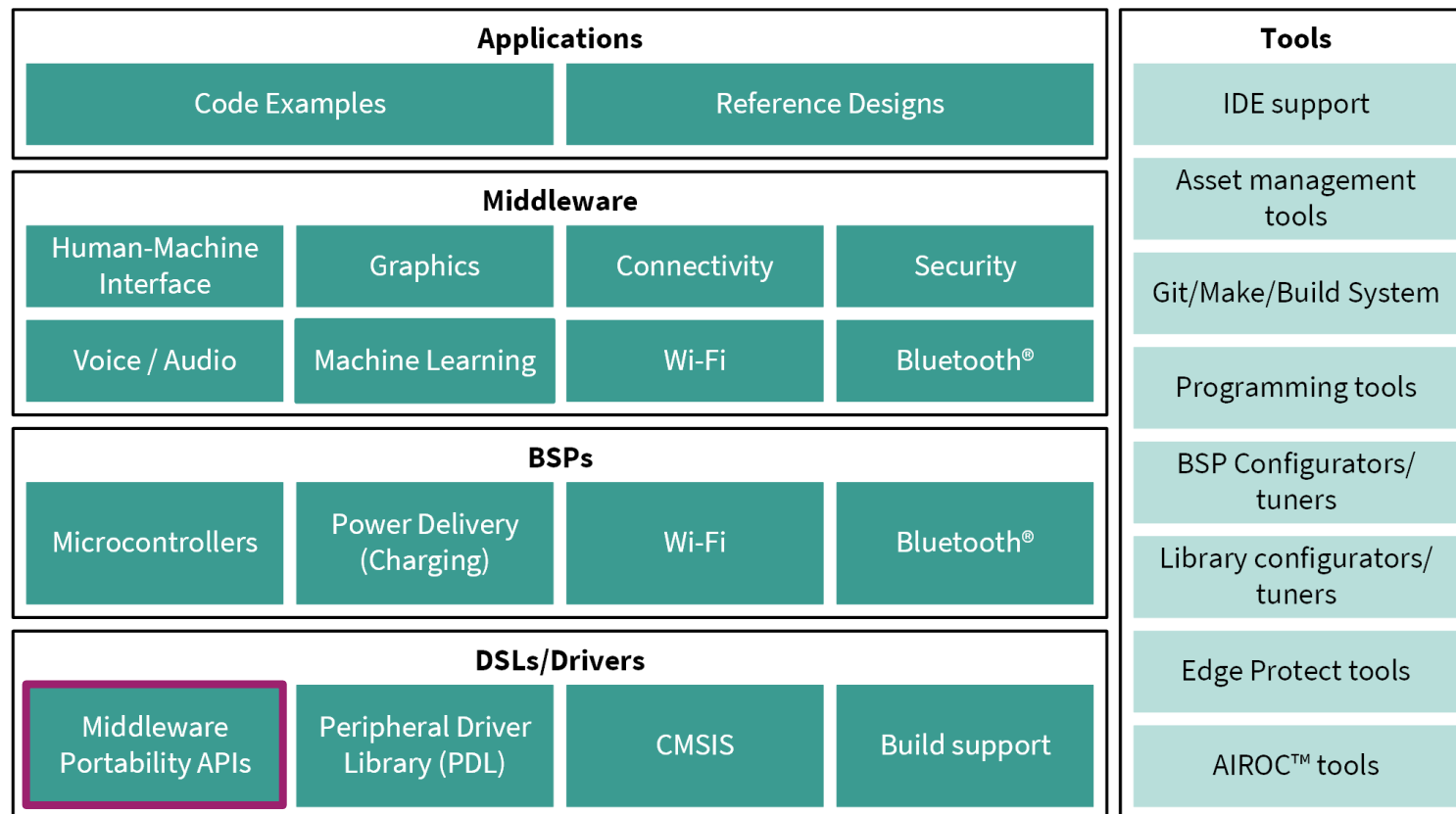
- Code examples
- Reference Designs

➤ Middleware Libraries



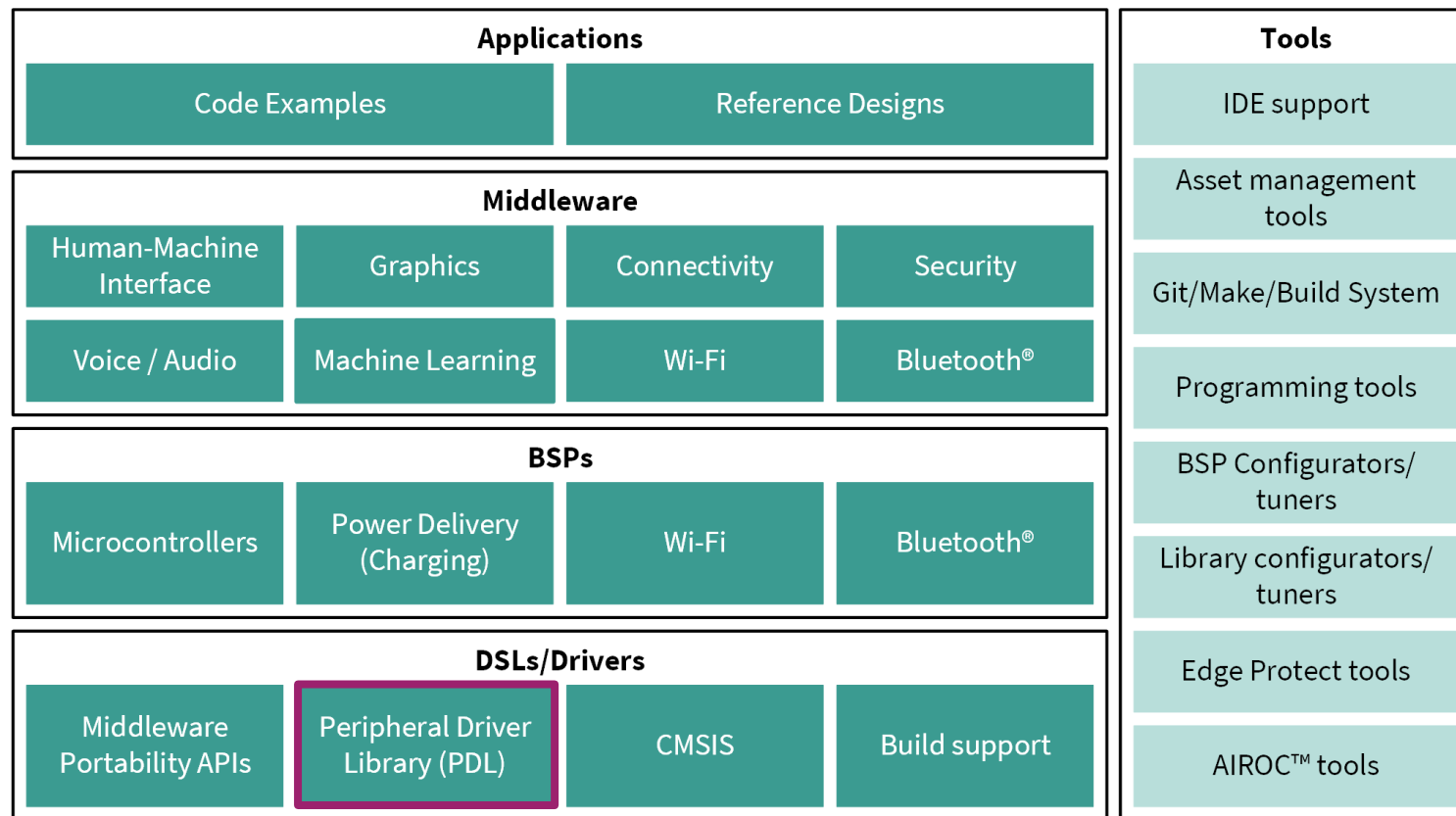
Firmware/application development : Software Resources

- Middleware Portability APIs
 - High-level APIs built on top of PDL
 - Hardware generic, so more portable
- Architected to enable Infineon maintained middleware to work across a wider range of devices
- Replacement of the former ModusToolbox HAL layer



Firmware/application development : Software Resources

- Peripheral Driver Library (PDL)
 - Set of low-level APIs
 - Hardware specific so less portable



Firmware/application development : Software Resources



<https://infineon.github.io/mtb-dsl-pse8xxgp/html/index.html>

infineon PSOC E8XXGP Device Support Library

Home PDL Getting Started API Reference

PSOC E8XXGP Device Support Library

- ModusToolbox™ PSE8xxGP Device Support Library
 - PDL Getting Started
 - API Reference
 - SYS PM Callback API Reference
 - HAL API Reference
 - PDL API Reference
 - NNLITE HW Accelerator KERNEL LIB

ModusToolbox™ PSE8xxGP Device Support Library (DSL)

Overview

The ModusToolbox™ Device Support Library provides all necessary device specific components to be used during software development for the PSE8xxGP family of devices.

Features

The package includes:

- PSOC™ Edge MCU Peripheral Driver Library (PDL) - provides, device header files, start-up code, and peripheral drivers as a set of low-level APIs to configure, initialize, and use various supported peripheral drivers
- Hardware Abstraction Layer (HAL) - provides high-level API wrappers around lower-level device specific APIs (such as PSOC™ MCU PDL)
- Device Utilities (device-utils) - handles the pre-defined System Power Management (SysPm) callback implementations for peripherals which require special handling when the MCU enters or exits DeepSleep
- make - provides build recipe make files and scripts for building and programming device applications

See the Release Notes for release-specific information.

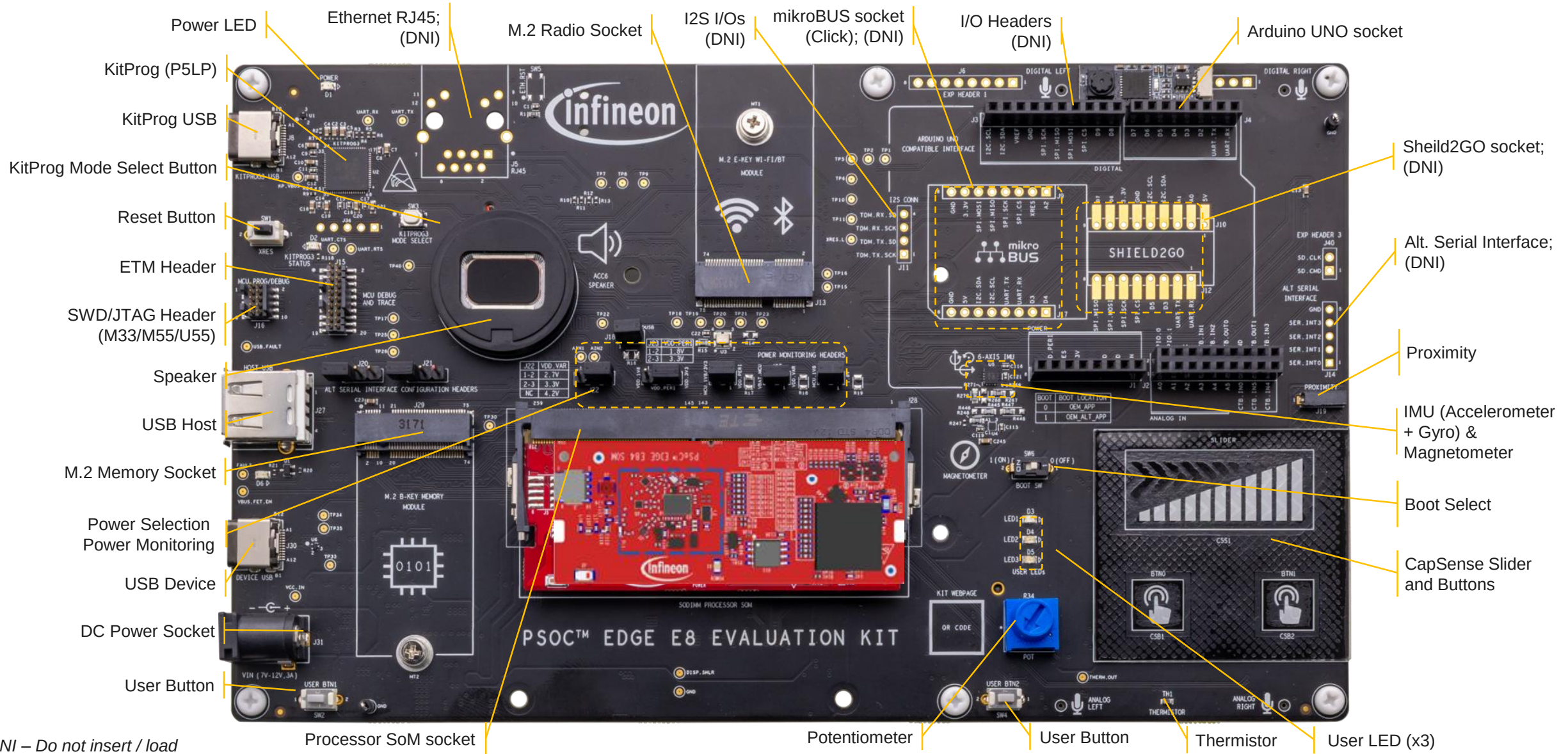
More information

Use the following links for more information, as needed:

- PSOC™ Edge Overview
- Device Support Library API Reference Manual
- Infineon GitHub
- ModusToolbox™

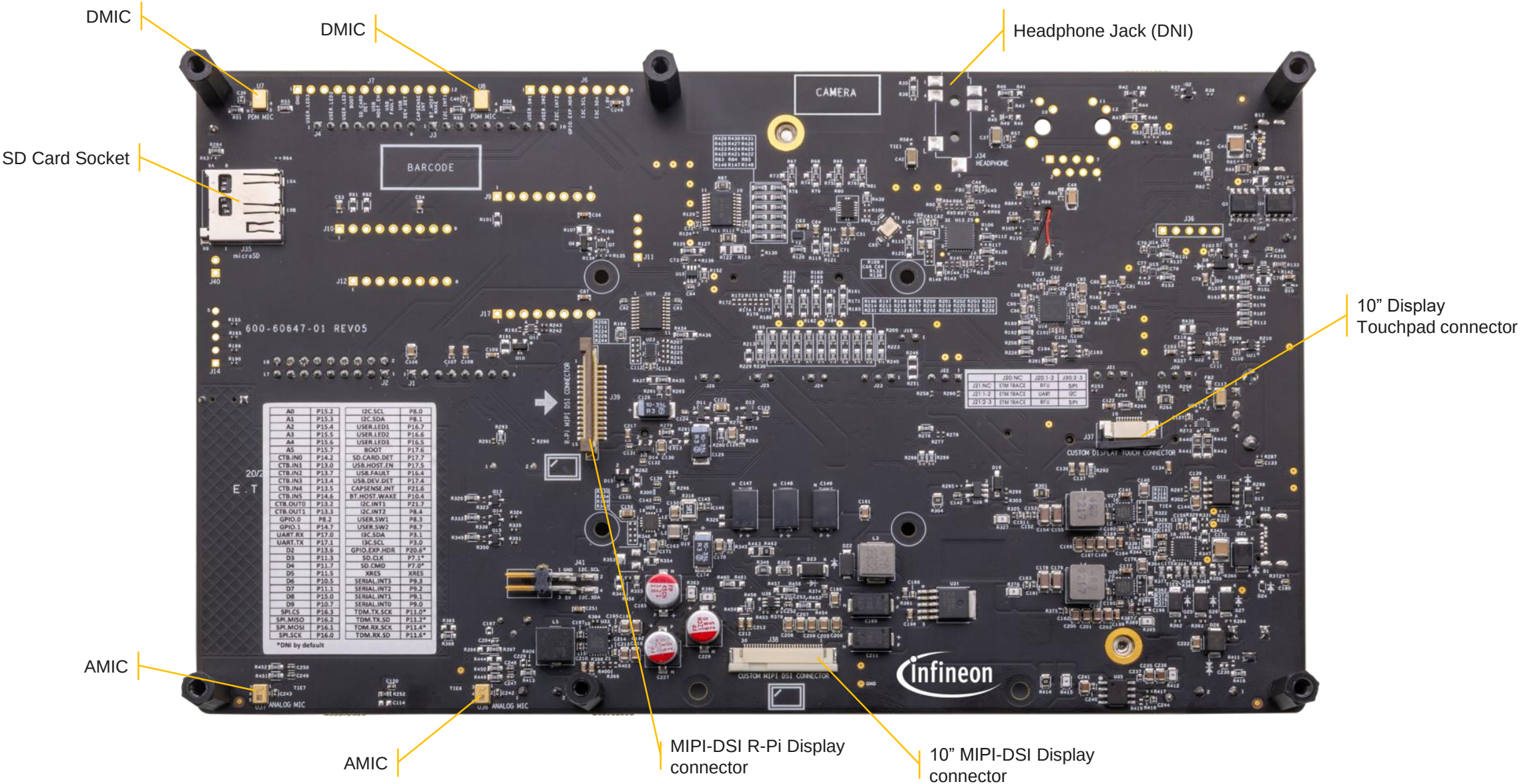
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PSOC™ Edge E84 MCU development kit – Top View

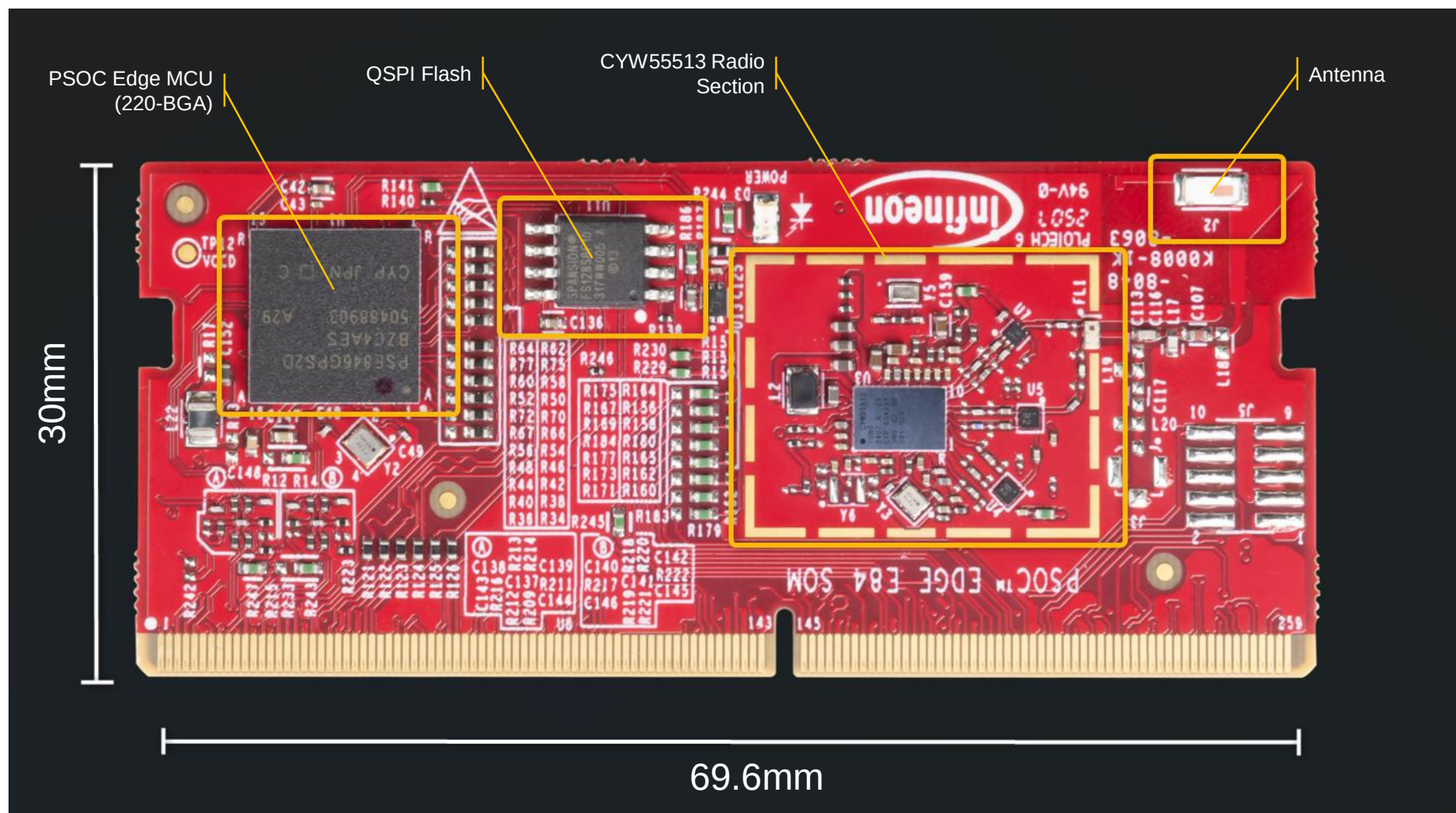


*DNI – Do not insert / load

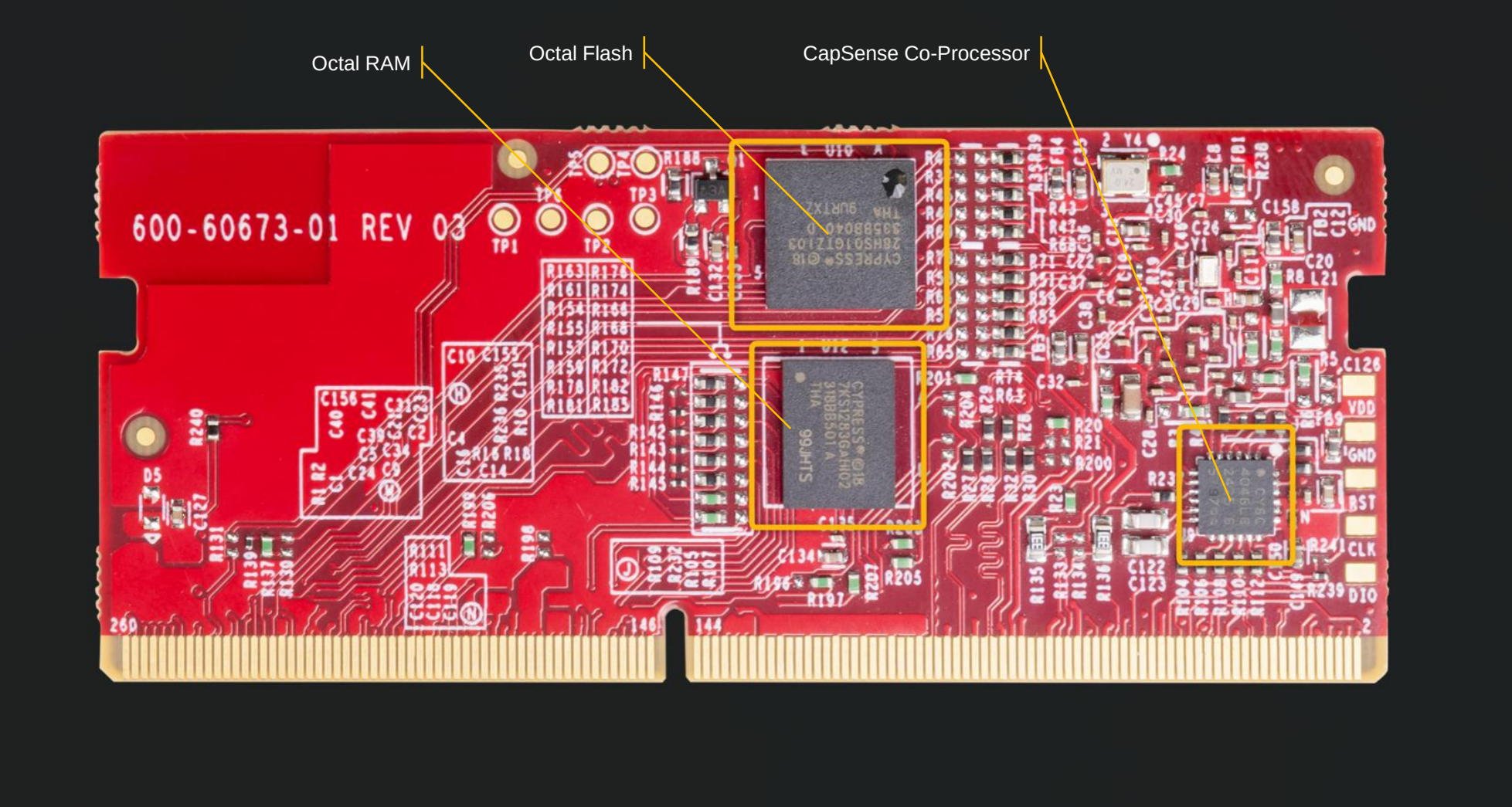
PSOC™ Edge E84 MCU development kit – Back View



PSOC™ Edge E84 MCU SOM – Top View



PSOC™ Edge E84 MCU SOM – Bottom View



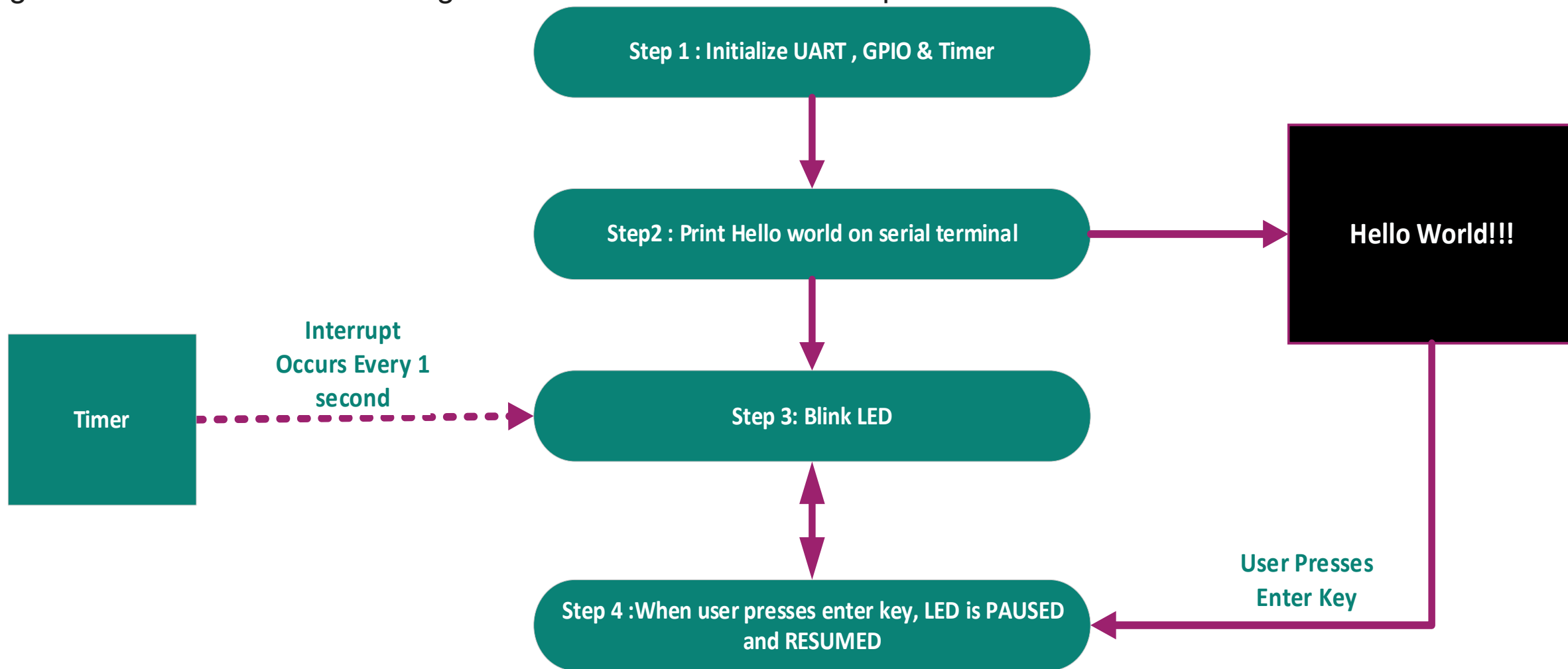
Programming/debugging

- Programming and Debugging PSOC™ Edge E84 MCU
 - Comes with a KitProg3 onboard programmer and debugger
 - Supports CMSIS-DAP
 - Supports OpenOCD using a GDB server and the J-Link debug probe
 - Debug environment is IDE specific – refer to MTB documentation.

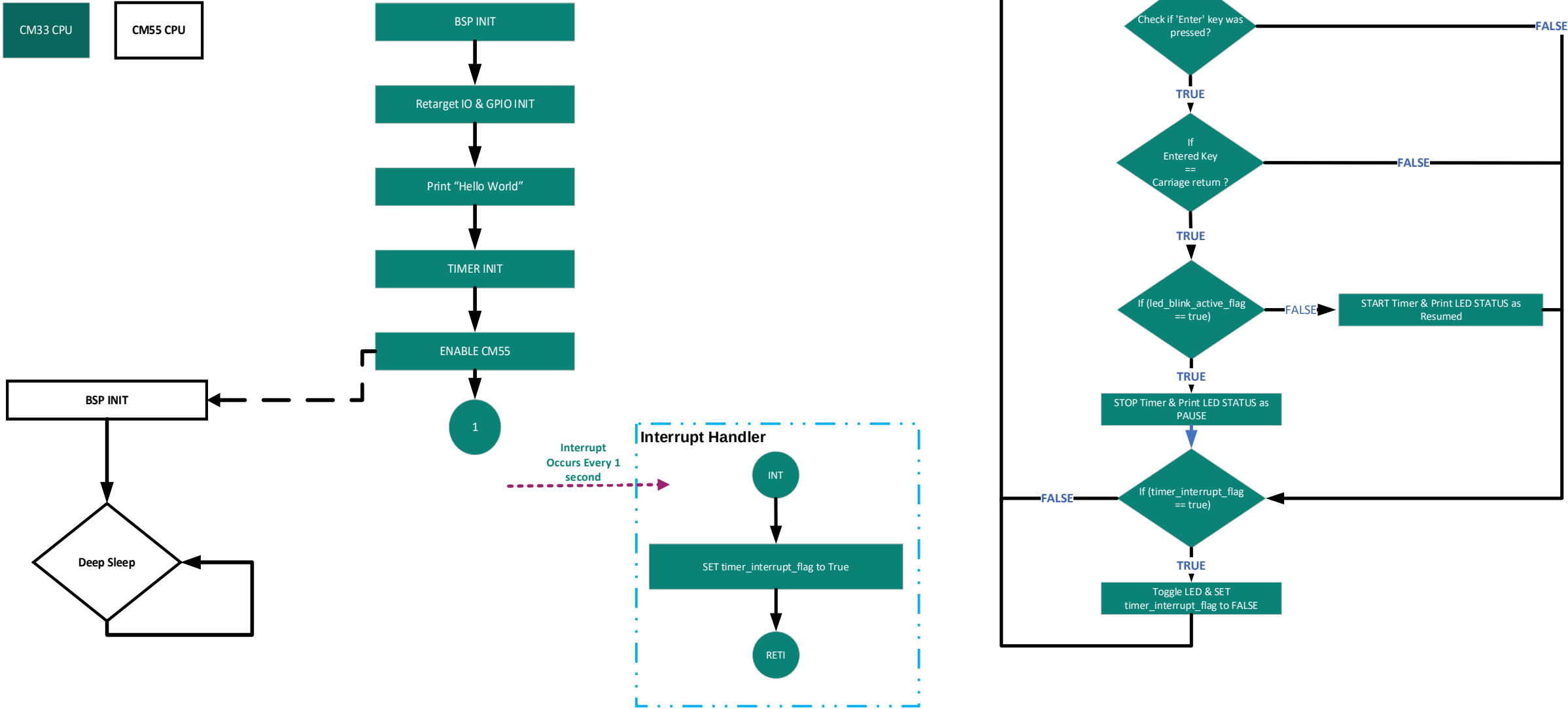
Code Example #1: Hello World + Blinky

Description of Hello World + Blinky example

Algorithm flow :This PSOC™ Edge E84 Hello World Code example demonstrates



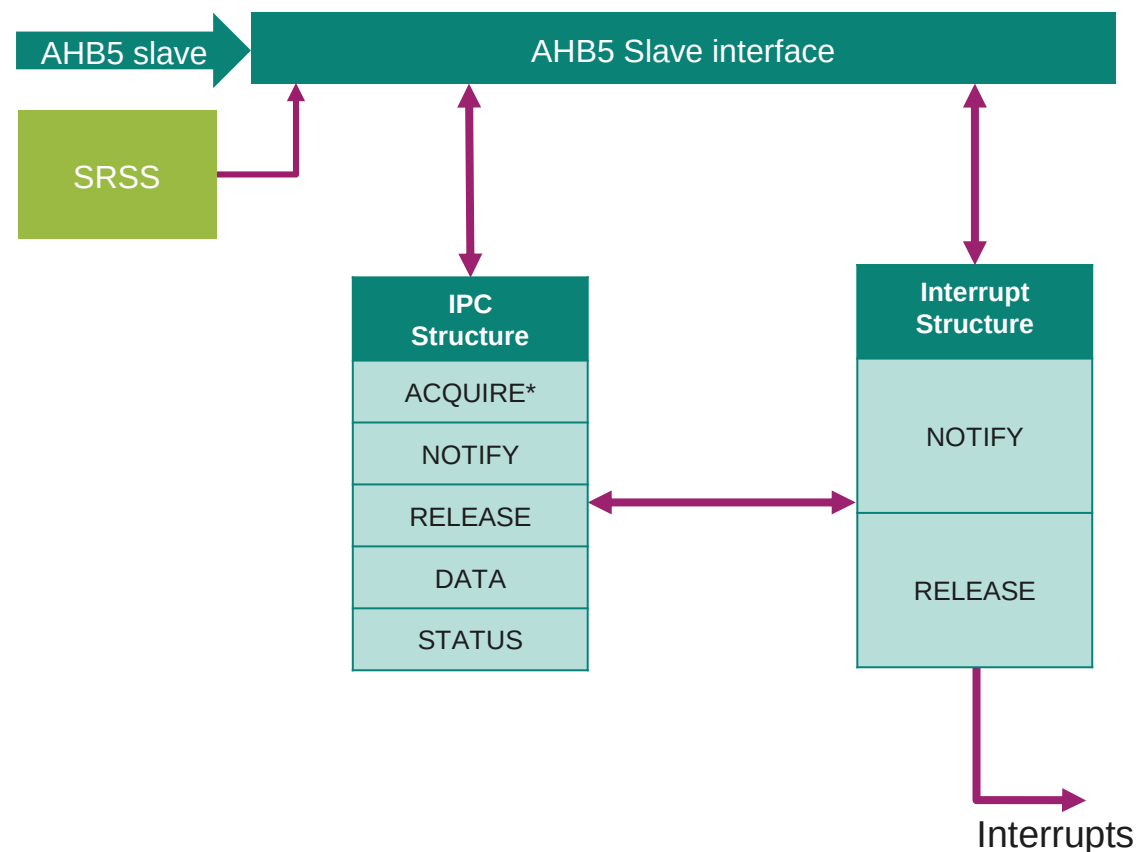
Flow chart



Code Example #2: IPC Semaphores using PDL

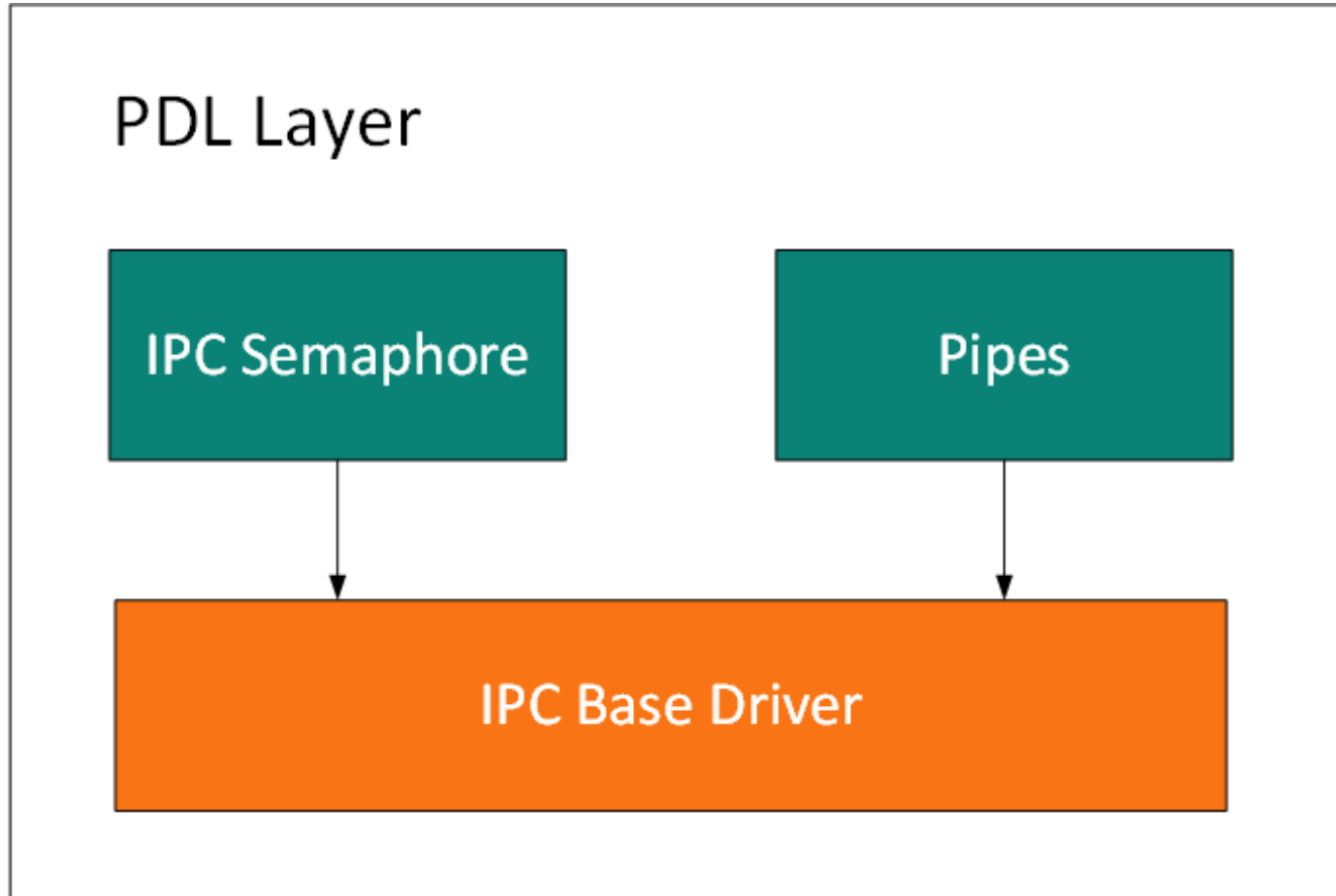
Overview of IPC

- Supports 16 communication channels
- Supports 16 interrupt structures
- Inter processor communication methods
 - Half-duplex with shared event handling
 - Half-duplex with independent event handling
 - Full-duplex communication
- IPC is available in CPU active power mode only.



* MS field of the Acquire register width has been expanded to 8 bits

Block Diagram of IPC PDL



IPC PDL (Peripheral Abstraction Layer) Functions

Functions

PDL API Reference » IPC (Inter Process Communication) » IPC semaphores layer (IPC_SEMA)

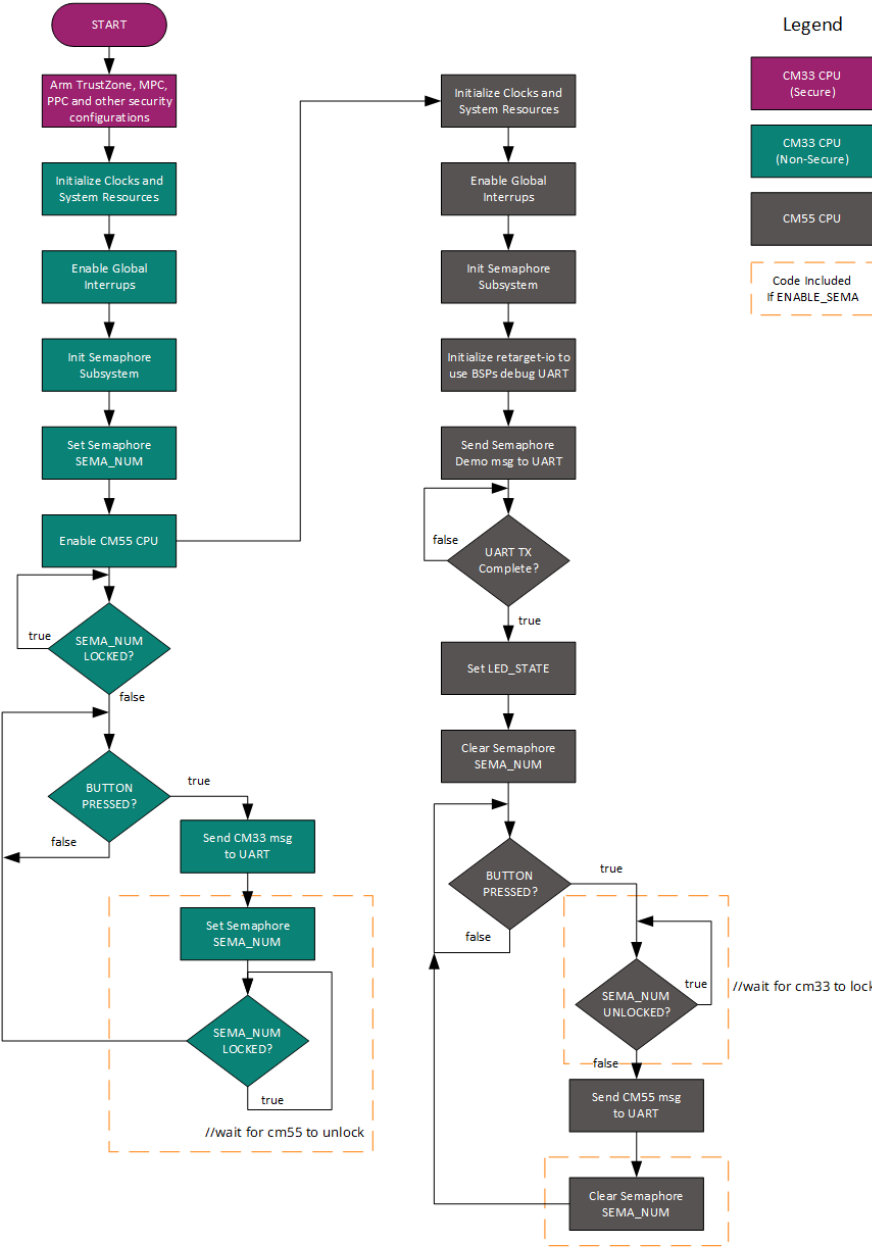
General Description

Functions are used in the driver.

Functions

<code>cy_en_ipcsema_status_t</code>	<code>Cy_IPC_Sema_Init</code> (uint32_t ipcChannel, uint32_t count, uint32_t memPtr[]) This function initializes the semaphores subsystem. More...
<code>cy_en_ipcsema_status_t</code>	<code>Cy_IPC_Sema_InitExt</code> (uint32_t ipcChannel, <code>cy_stc_ipc_sema_t</code> *ipcSema) This function initializes the semaphores subsystem. More...
<code>cy_en_ipcsema_status_t</code>	<code>Cy_IPC_Sema_Set</code> (uint32_t semaNumber, bool preemptable) This function tries to acquire a semaphore. More...
<code>cy_en_ipcsema_status_t</code>	<code>Cy_IPC_Sema_Clear</code> (uint32_t semaNumber, bool preemptable) This functions tries to releases a semaphore. More...
<code>cy_en_ipcsema_status_t</code>	<code>Cy_IPC_Sema_Status</code> (uint32_t semaNumber) This function returns the status of the semaphore. More...
<code>uint32_t</code>	<code>Cy_IPC_Sema_GetMaxSems</code> (void) This function returns the number of semaphores in the semaphores subsystem. More...

Flow chart





Thank you! Stay tuned for more trainings...

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